You need to:

- **Understand notation of Venn diagrams.**
- **Definition of sets.**
- **Use language, notation and Venn diagrams to describe sets and represent relationships between sets. (Extended)**

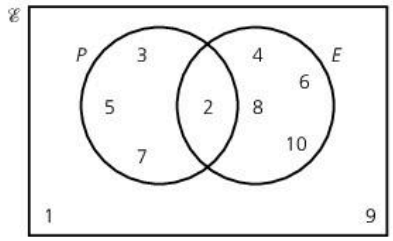
A **set** is a collection of items. These may be numbers, people, letters, etc.

You use 'curly' brackets to represent sets.

For example, the set P, of prime numbers less than or equal to 10, can be represented as $P = \{2, 3, 5, 7\}$.

The number of **elements** in set P is denoted by $n(P)$, so in this case $n(P) = 4$.

A **Venn diagram** can be used to represent sets.



In the Venn diagram, P is the set of prime numbers less than or equal to 10 and E is the set of even numbers less than or equal to 10.

The symbol $\mathscr{E}$ is the symbol for the **universal set**. This is the set of all things that are being considered at the time, in this case, the set of integers from 1 to 10.

The element '2' lies inside both circles. This region of the Venn diagram is known as the **intersection** and contains all of the elements that are in **both** sets.

Recap

The symbol for intersection is $\cap$.

In the Venn diagram above:

$P \cap E = \{2\}$

$n(P \cap E) = 1$

Elements which lie in either P or E or both lie in the **union** of P and E.

Recap

The symbol for union is $\cup$.

In the Venn diagram above:

$P \cup E = \{2, 3, 4, 5, 6, 7, 8, 10\}$

$n(P \cup E) = 8$

Elements that do not lie in P are called **complementary**.

Recap

The symbol for the complement of set P is P'.

In the Venn diagram above:

$P' = \{1, 4, 6, 8, 9, 10\}$

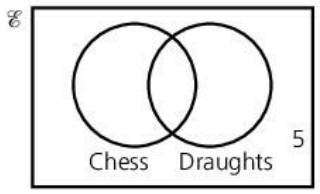
$n(P') = 6$

Worked example

In a class of 33 students, 20 like chess, 12 like draughts and 5 like neither.

(a) Represent this information in a Venn diagram. **[3 marks]**
(b) How many students like only one of chess or draughts? **[1 mark]**

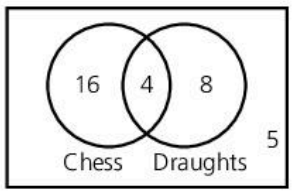
(a)



There are 28 students left.

There are 20 who like chess and 12 who like draughts.

$20 + 12 - 28 = 4$



(b) The number of students who like only one of chess or draughts is $16 + 8 = 24$

Exam tip

Draw a Venn diagram with two circles overlapping and add in the 5 students who like neither game outside of both circles.

Exam tip

Add up the number of students who like chess and who like draughts then subtract the number of students left to find the number of students who like both.

Fill in the 4 in the intersection. $20 - 4 = 16$ students like chess only. $12 - 4 = 8$ students like draughts only.

Extended

Recap

You need to know several other symbols and notations for sets.

$\in$ means '... is an element of...'

$\notin$ means '... is not an element of...'

$\varnothing$ means the empty set or $\{\}$

$A \subseteq B$ means 'A is a subset of B' (and A *can* equal B)

$A \not\subseteq B$ means 'A is not a subset of B'

$A \subset B$ means 'A is a proper subset of B' (and A *cannot* equal B)

$A \not\subset B$ means 'A is not a proper subset of B'

Apply

$A = \{\text{Integers from 1 to 10}\}$

$B = \{\text{Odd numbers between 0 and 20}\}$

$C = \{2, 4, 6, 8, 10\}$

Which of the following statements are true and which are false?

- $2 \in C$
- $4 \in B$
- $8 \notin A$
- $14 \notin B$
- $A \subseteq B$
- $C \subset A$
- $B \cap C = \varnothing$

Extended

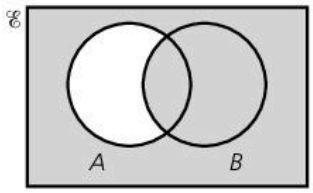
Worked example

Draw Venn diagrams and shade the region(s) that represent:

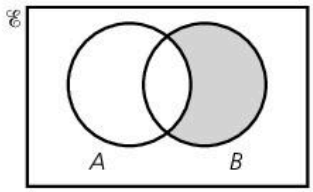
(a) $A' \cup B$

(b) $A' \cap B$ **[2 marks]**

(a)



(b)

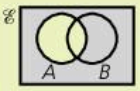
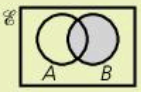


Key skills

You need to be able to shade Venn diagrams to represent different situations.

Exam tip

Imagine shading the region A' and the region B separately.

A':  B: 

The union is any region that is shaded in either diagram.

The intersection is any region that is shaded in *both* of the separate diagrams.

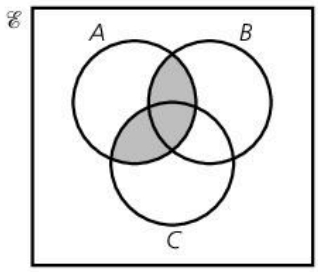
Worked example

Draw Venn diagrams and shade the region(s) that represent:

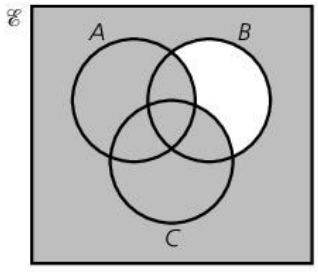
(a) $(B \cup C) \cap A$

(b) $(A \cup C) \cup B'$ **[2 marks]**

(a)



(b)



Key skills

You need to be able to work with Venn diagrams containing three circles.

Exam tip

Look for the regions that lie in B or C and **also** in A.

Exam tip

The only region **not** shaded is the one that lies in B, but not in A or C.

Questions

1 In a group of 100 students, 70 enjoy Maths, 50 enjoy French and 20 enjoy neither.

a Draw a Venn diagram showing this information.

b Use your diagram to find the number of students who enjoy both subjects.

2 On an athletics day 150 athletes take part. 60 are in the 100 metres race, 50 are in the 200 metres race and 80 are in neither.

a Draw a Venn diagram showing this information.

b Use the diagram to find the number of athletes who ran in only one race.

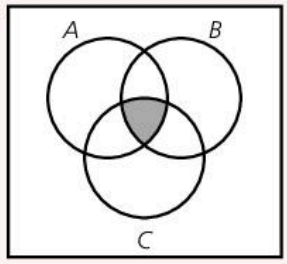
3 In a shop there were 120 customers on a certain day. Of these, 60 paid using notes, 30 paid using coins and 50 paid using cards. There were no customers who paid using both cards and cash.

a Draw a Venn diagram showing this information.

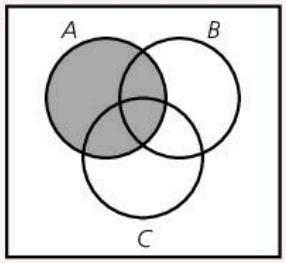
b Use your diagram to find the number of customers who used both notes and coins.

4 Describe the shaded regions:

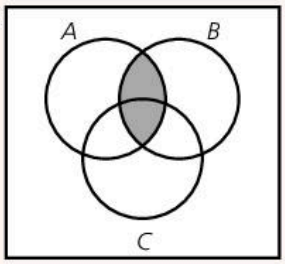
a



b



c



d

